

# Convergence and performance results for reinforcement learning with two time scales

Diogo Filipe de Sousa Carvalho

Thesis proposal



# Outline

Reinforcement learning



Function approximation



Research problem

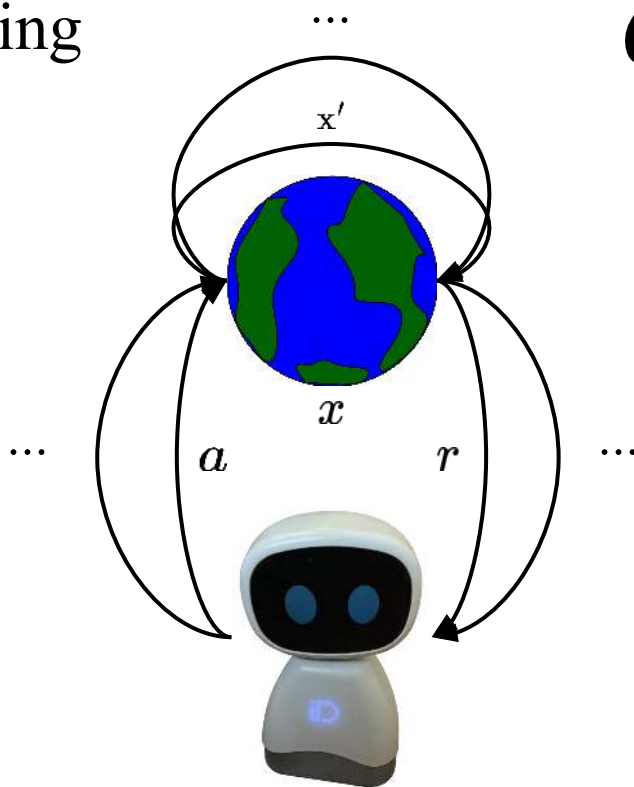


Expected contributions



Discussion

# Reinforcement learning



$$q^* = \mathbf{H}q^*$$

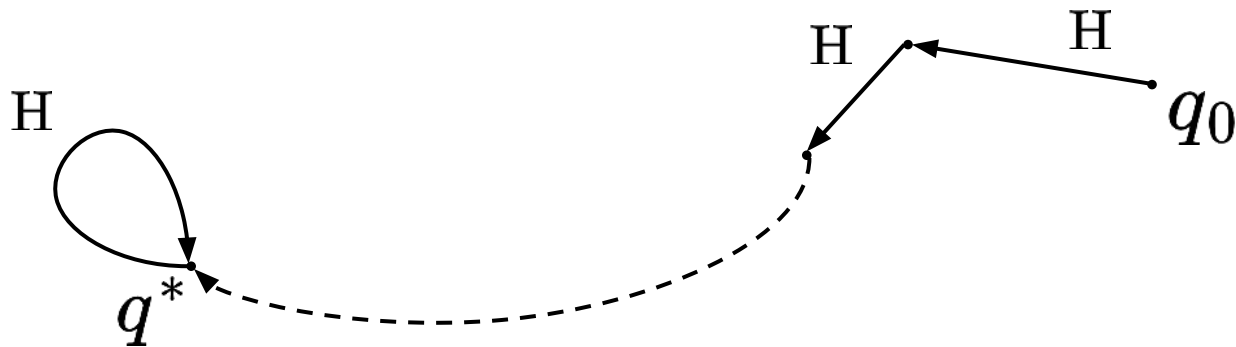
Bellman operator

discount factor

$$(\mathbf{H}q)(x, a) = \mathbb{E} [r + \gamma \cdot \max_{a'} q(x', a')]$$

# Reinforcement learning

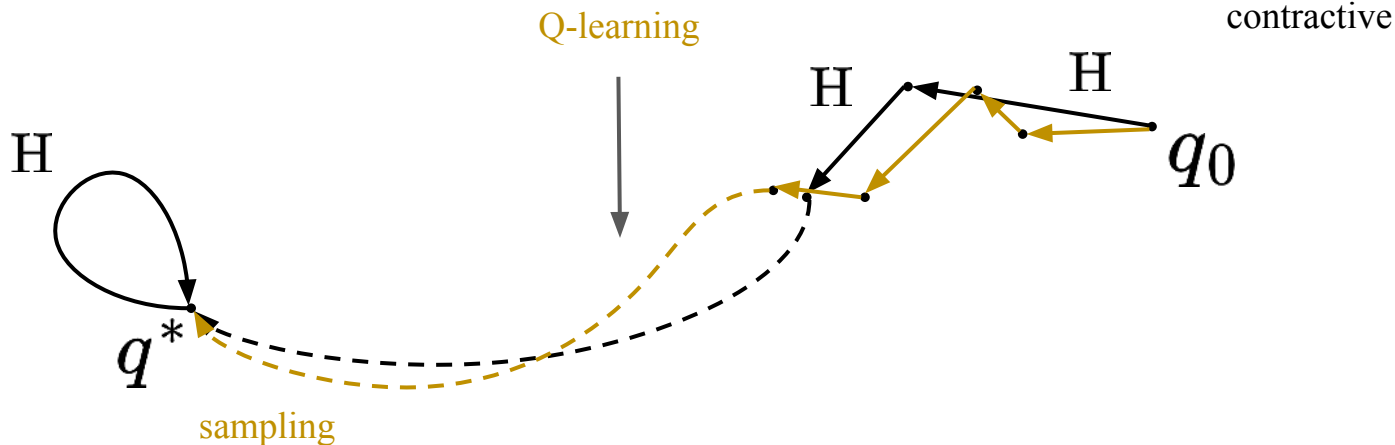
$$q^* = \mathbf{H}q^*$$



$$(\mathbf{H}q)(x, a) = \mathbb{E} [r + \gamma \cdot \max_{a'} q(\mathbf{x}', a')]$$

# Reinforcement learning

$$q^* = \mathbf{H}q^*$$

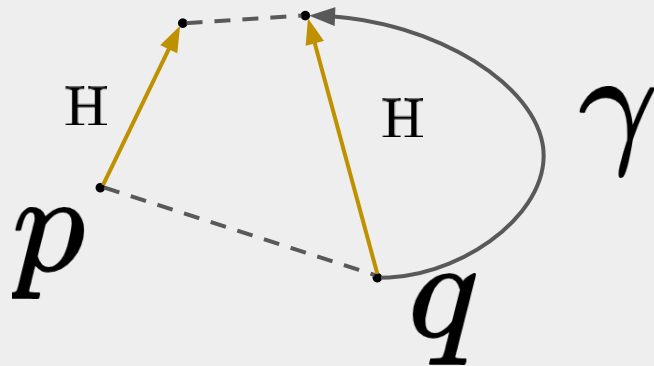


$$\mathcal{D} = \{(x_i, a_i, x'_i, r_i), i \in \mathbb{N}_0\}$$

$$(\mathbf{H}q)(x, a) = \mathbb{E}[r + \gamma \cdot \max_{a'} q(x', a')]$$

## Contractivity

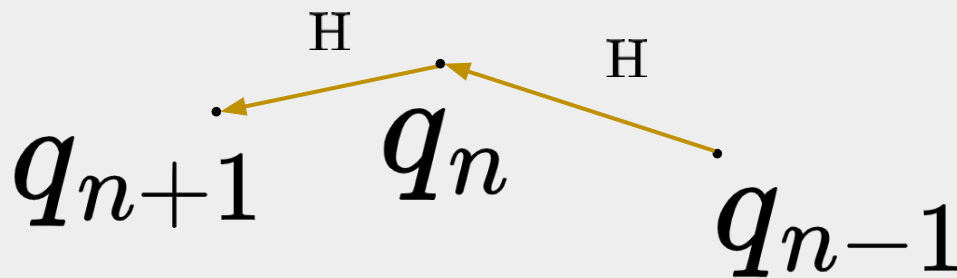
$$q^* = \mathbf{H}q^*$$



$$\|\mathbf{H}p - \mathbf{H}q\| \leq \gamma \cdot \|p - q\|$$

## Contractivity

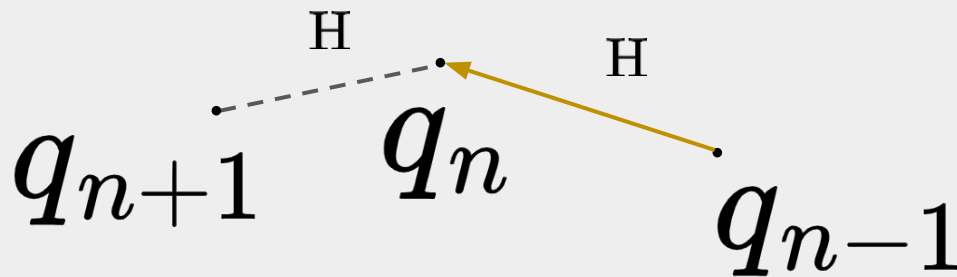
$$q^* = \mathbf{H}q^*$$



$$\|\mathbf{H}p - \mathbf{H}q\| \leq \gamma \cdot \|p - q\|$$

## Contractivity

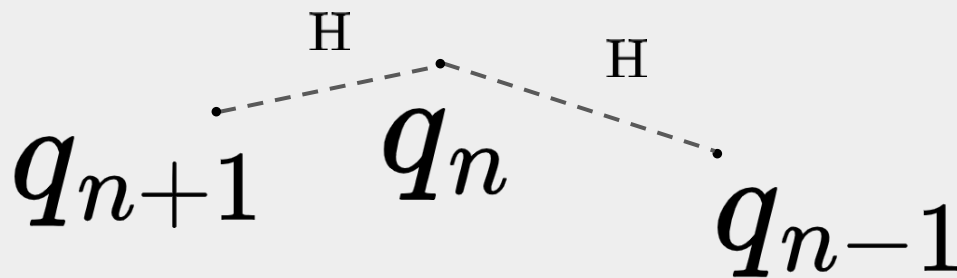
$$q^* = \mathbf{H}q^*$$



$$\|\mathbf{H}p - \mathbf{H}q\| \leq \gamma \cdot \|p - q\|$$

## Contractivity

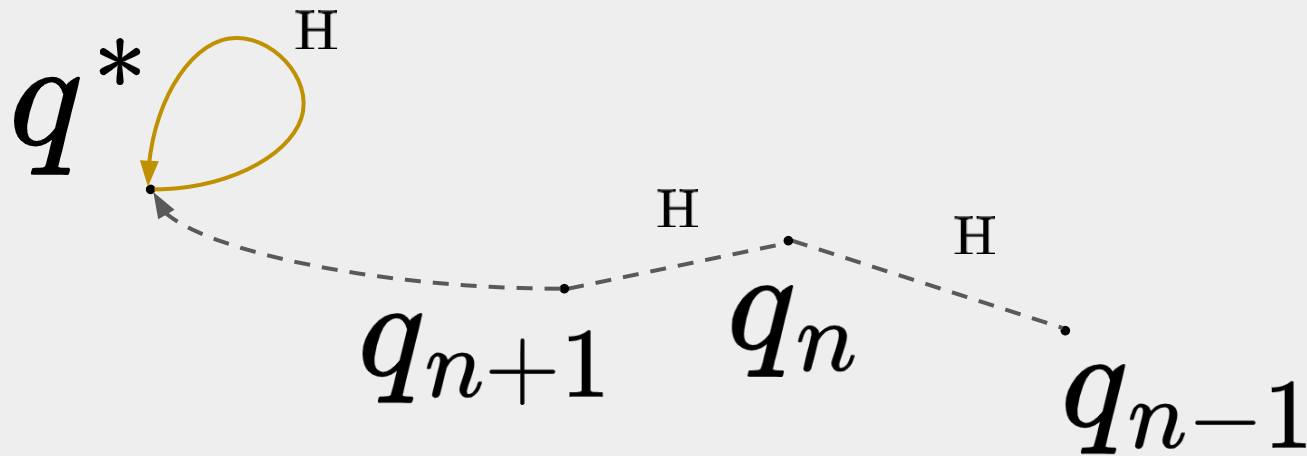
$$q^* = \mathbf{H}q^*$$



$$\|\mathbf{H}p - \mathbf{H}q\| \leq \gamma \cdot \|p - q\|$$

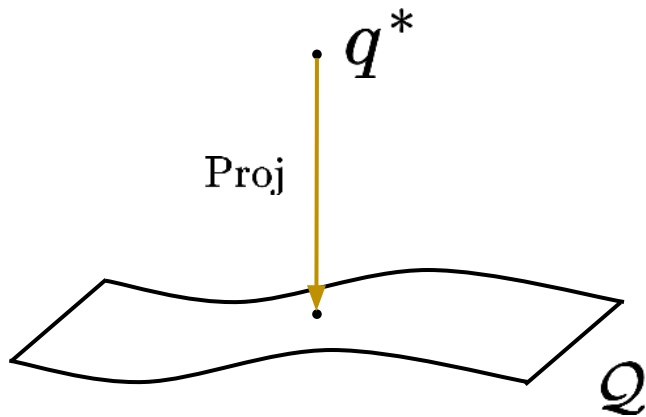
Contractivity

$$q^* = Hq^*$$



$$\|Hp - Hq\| \leq \gamma \cdot \|p - q\|$$

# Function approximation

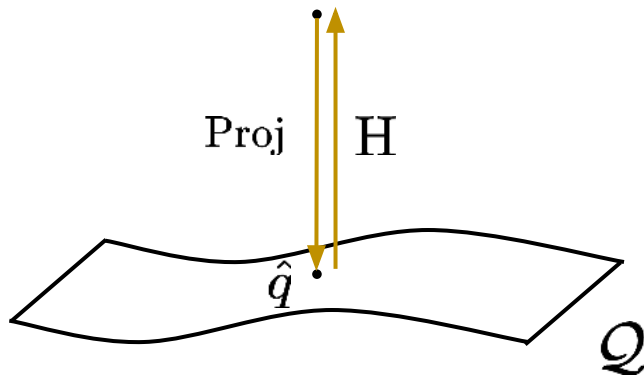


$$\text{Proj } q = \operatorname{argmin}_{\bar{q} \in Q} \|q - \bar{q}\|$$

$$Q = \{q_w, w \in \mathcal{R}^k\}$$

Function approximation

$$\hat{q} = (\text{Proj } H)\hat{q}$$



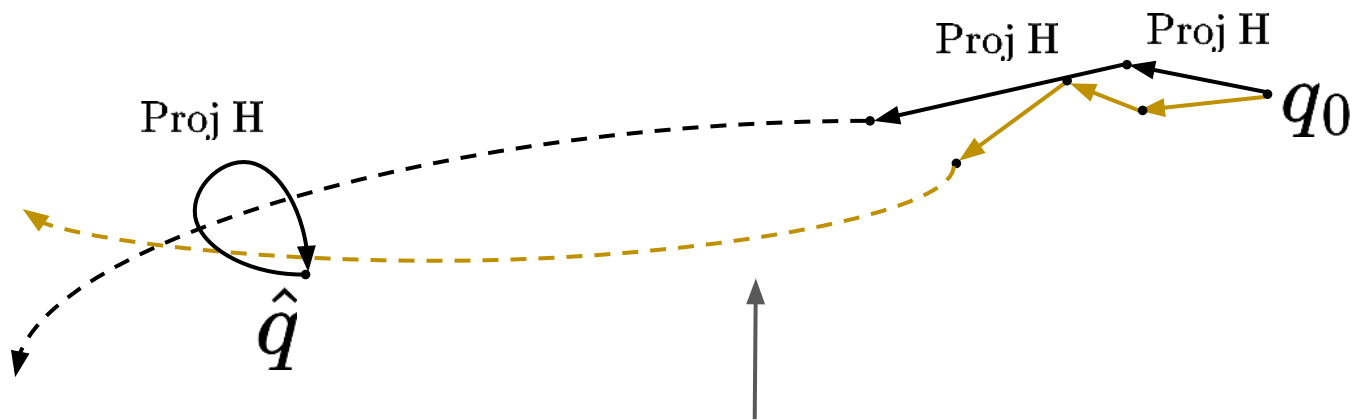
$$\text{Proj } q = \operatorname{argmin}_{\bar{q} \in \mathcal{Q}} \|q - \bar{q}\|$$

$$\mathcal{Q} = \{q_w, w \in \mathcal{R}^k\}$$

# Function approximation

$$\hat{q} = (\text{Proj } H) \hat{q}$$

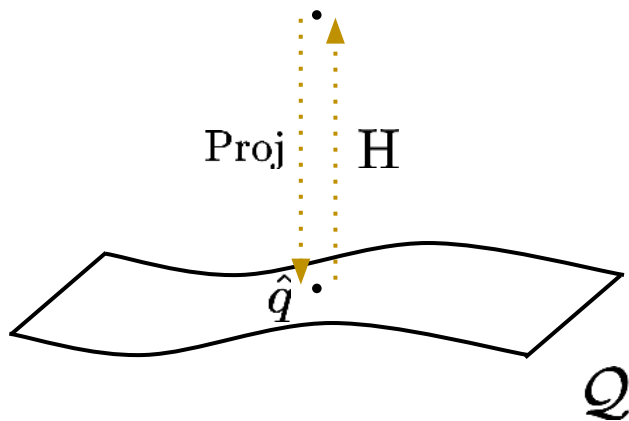
not contractive



Q-learning with  
function approximation

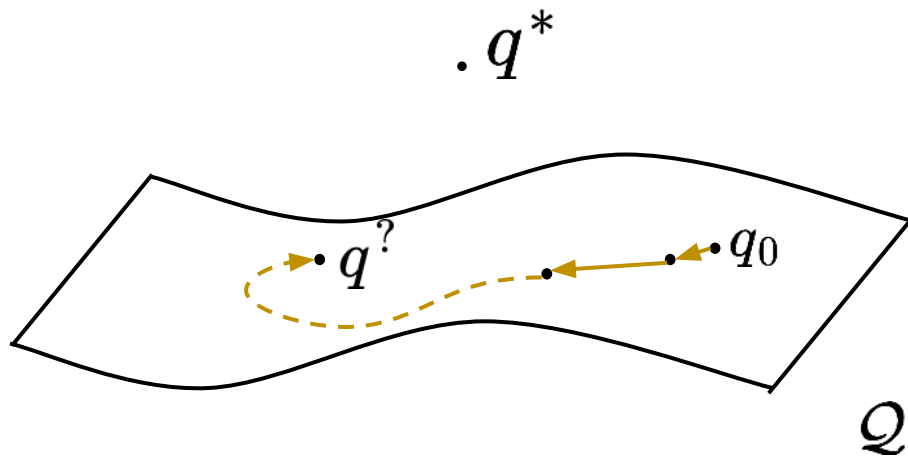
Function approximation

$$\hat{q} = (\text{Proj } H)\hat{q}$$



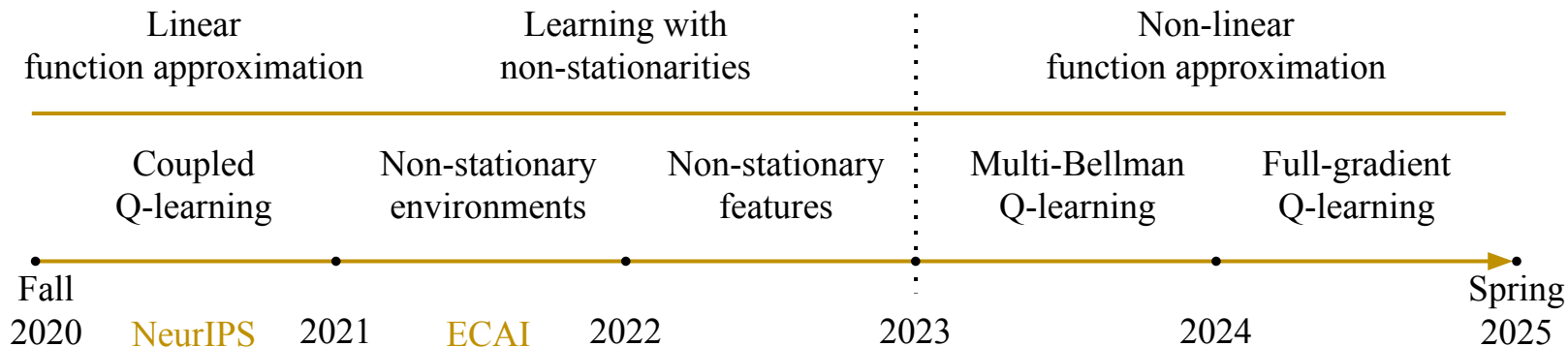
# Research problem

How can we approximate the optimal value function?

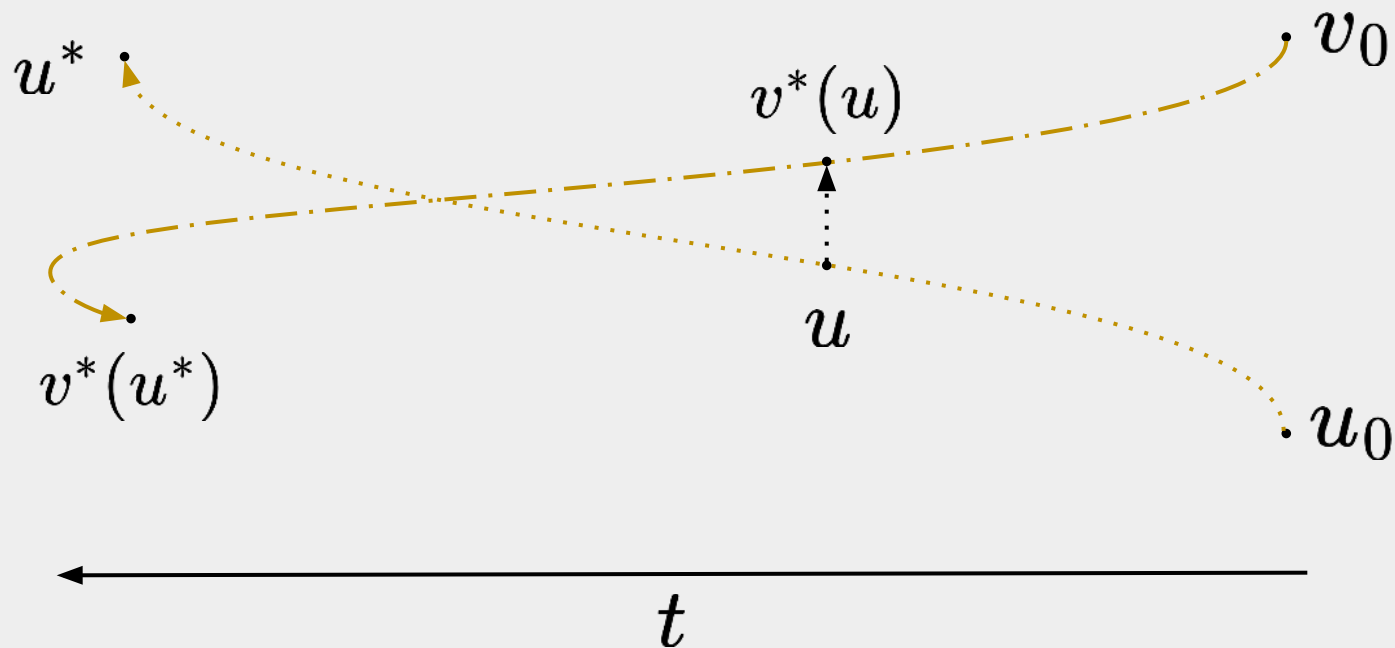


# Expected contributions

How can we approximate the optimal value function?

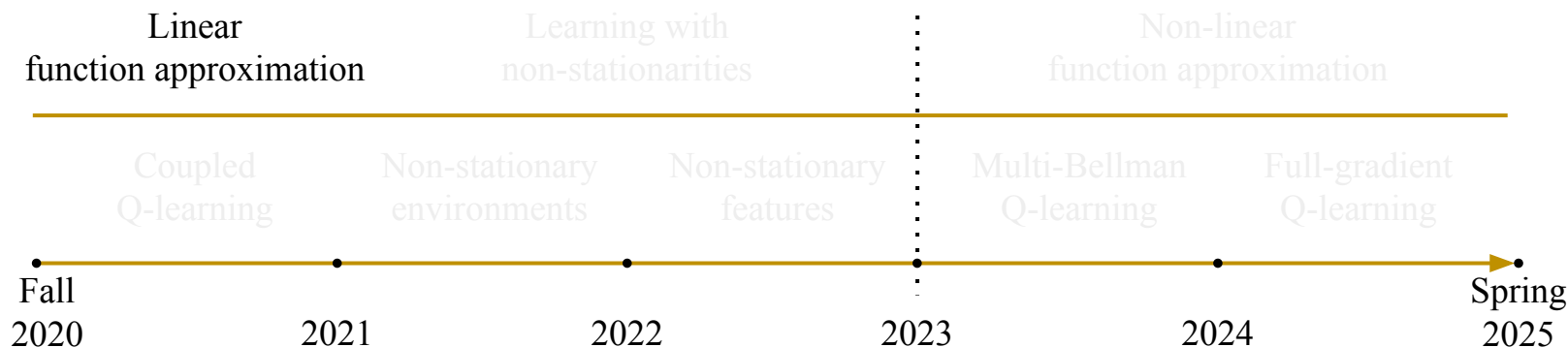


## Two time-scale stochastic approximation



# Expected contributions

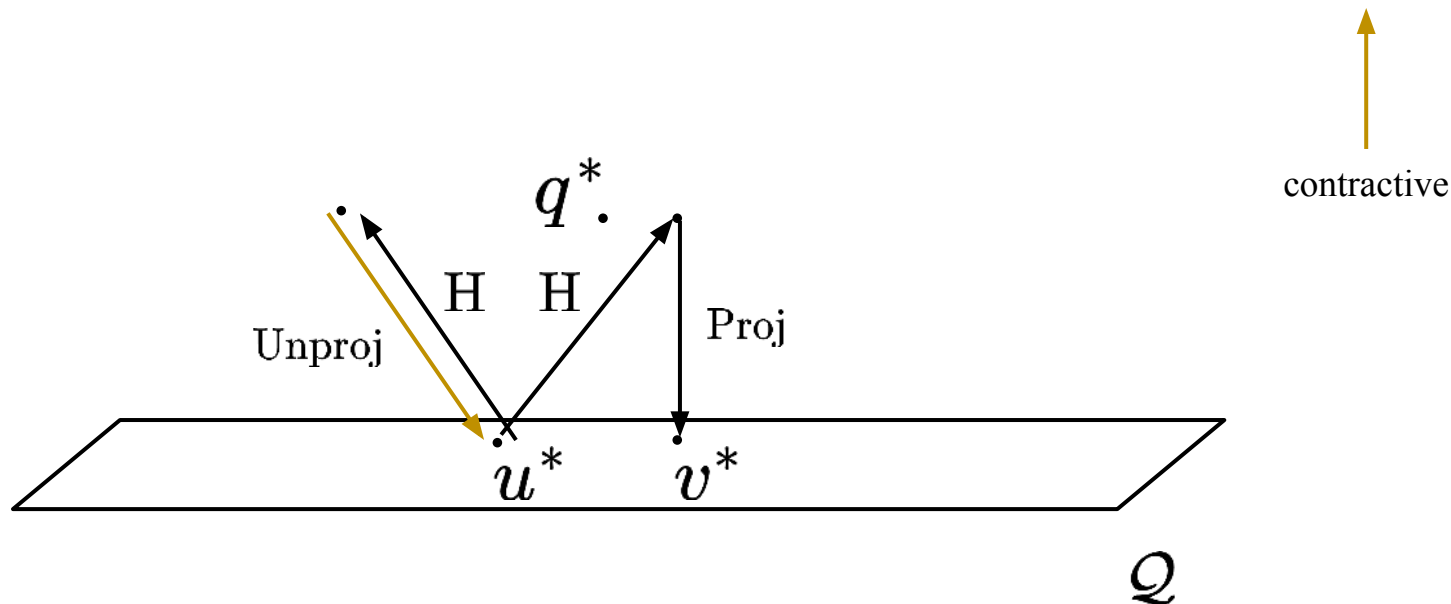
How can we approximate the optimal value function?



# Coupled Q-learning

$$v^* = (\text{Proj } H)u^*$$

$$u^* = (\text{Unproj } H)v^*$$



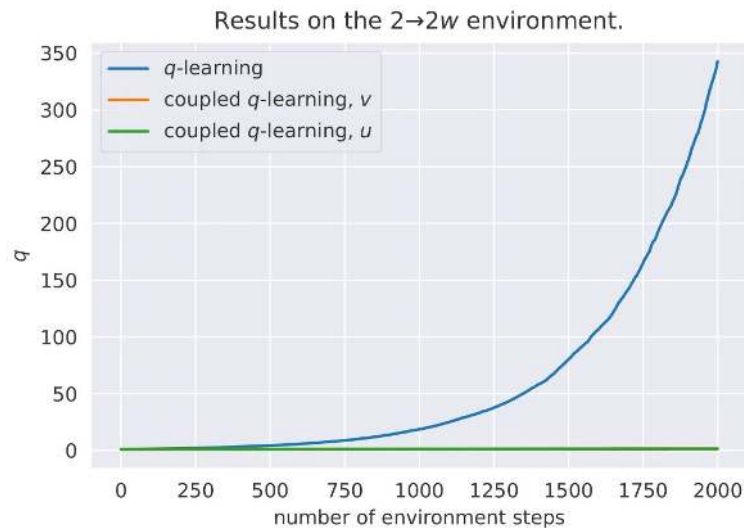
$$\mathcal{Q} = \{q_w : q_w(x, a) = \phi(x, a) \cdot w, w \in \mathbb{R}^k\}$$

# Coupled Q-learning



$$q^* = 0$$

# Coupled Q-learning



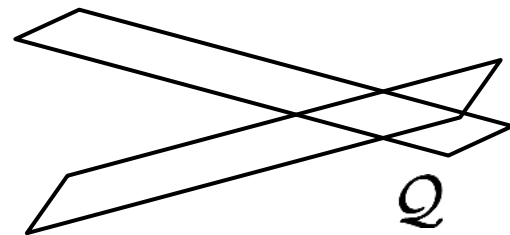
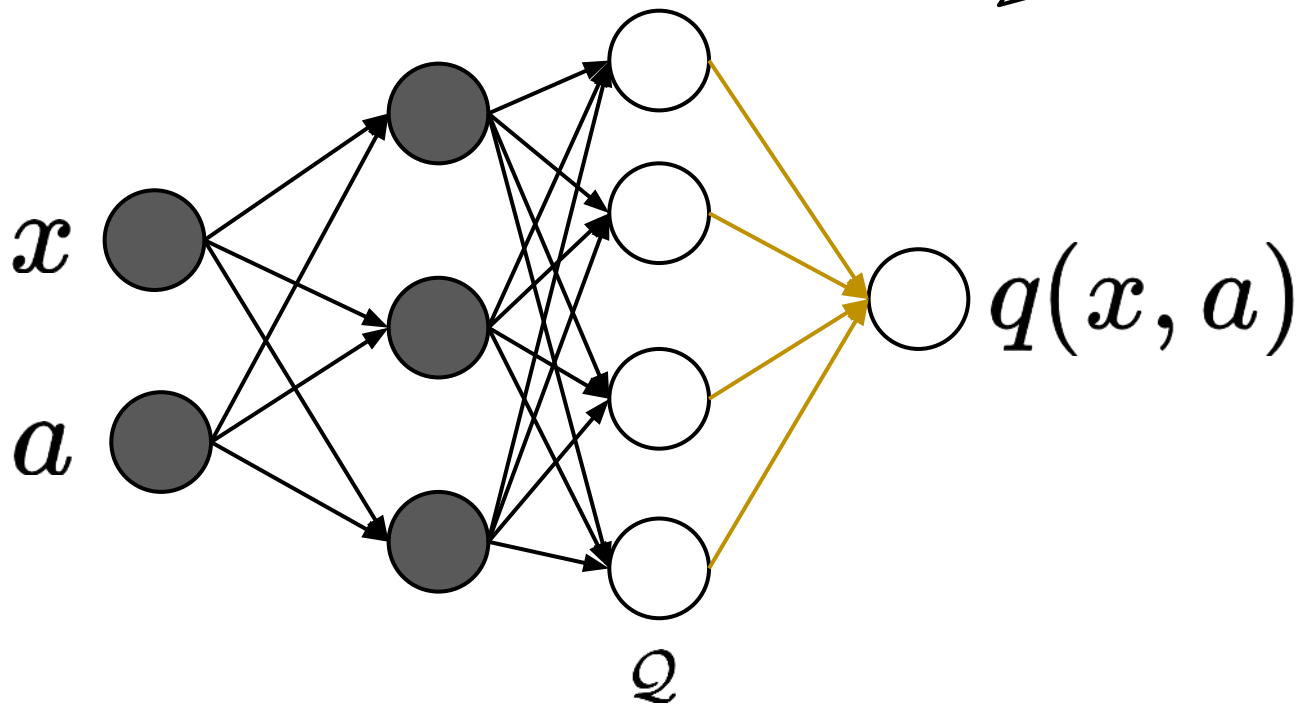
←  $q^*$

fixed  
features



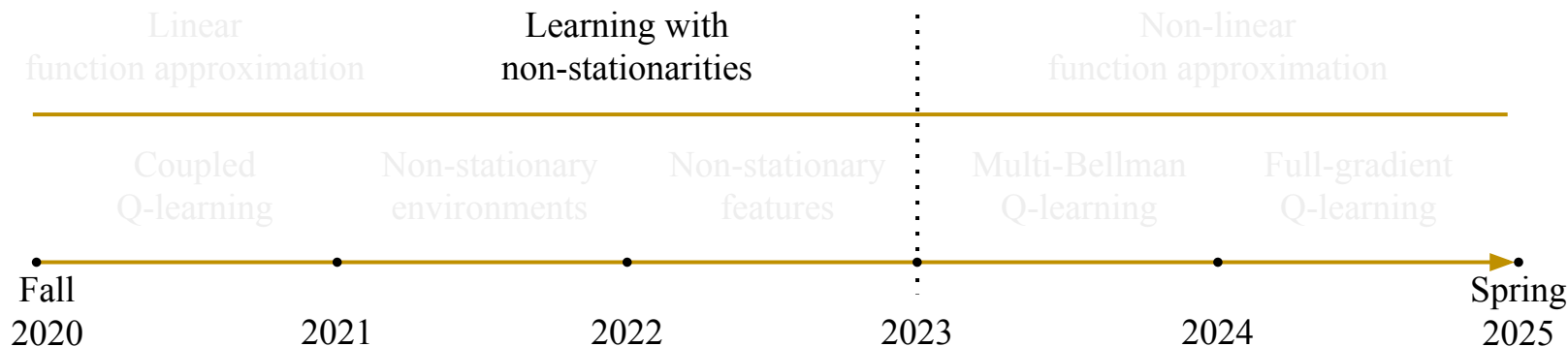
$Q$

# Deep reinforcement learning

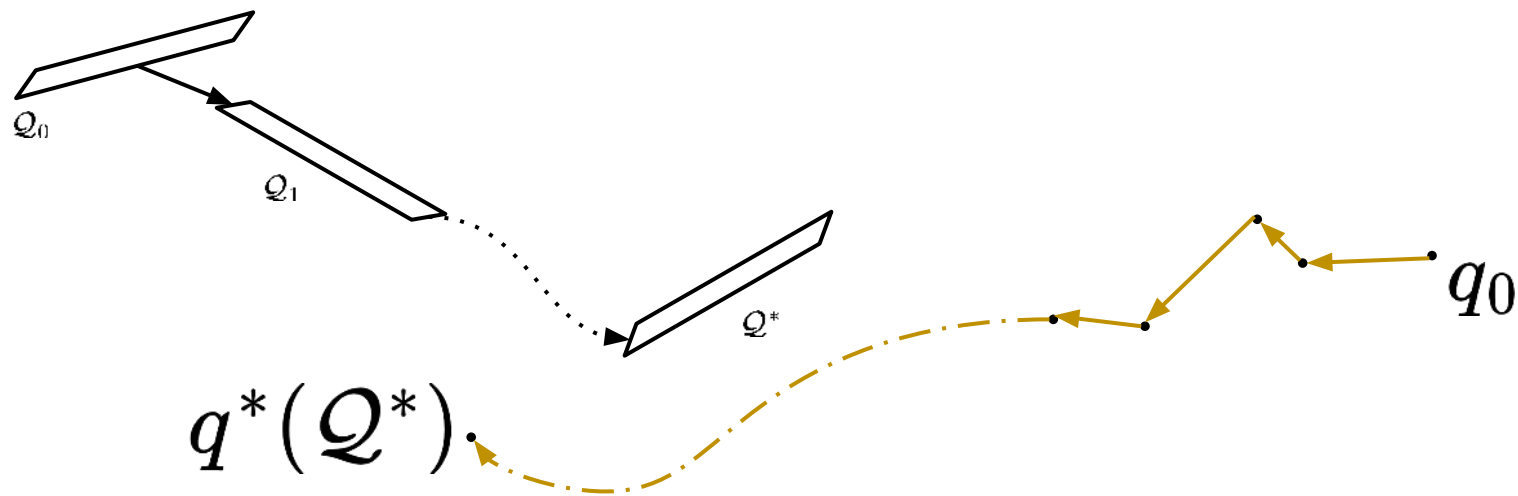


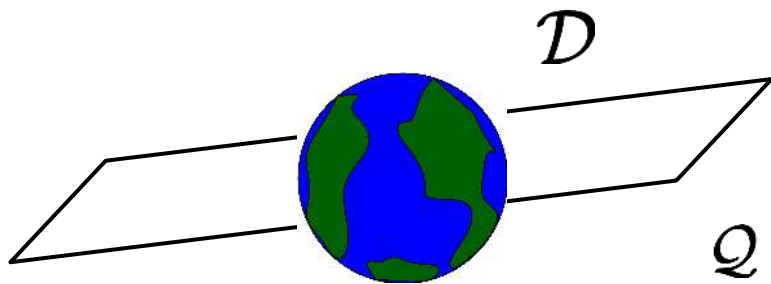
# Expected contributions

How can we approximate the optimal value function?

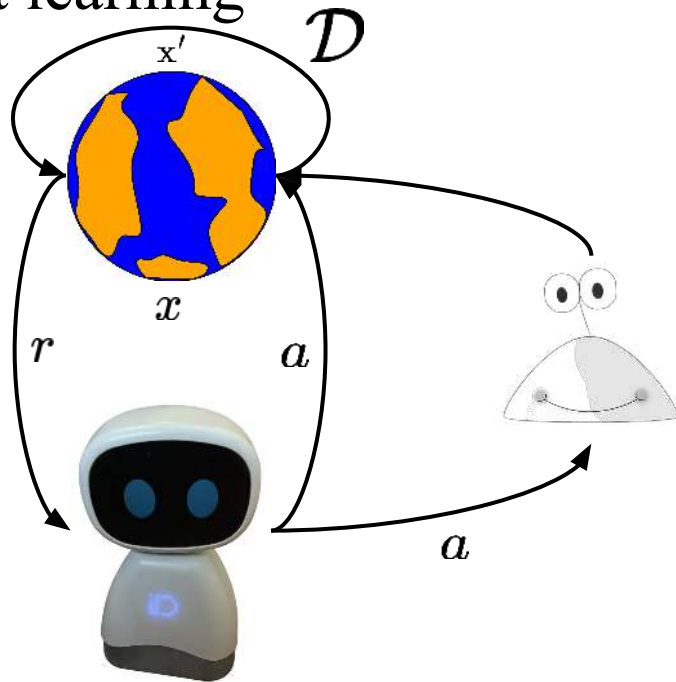


# Non-stationary features

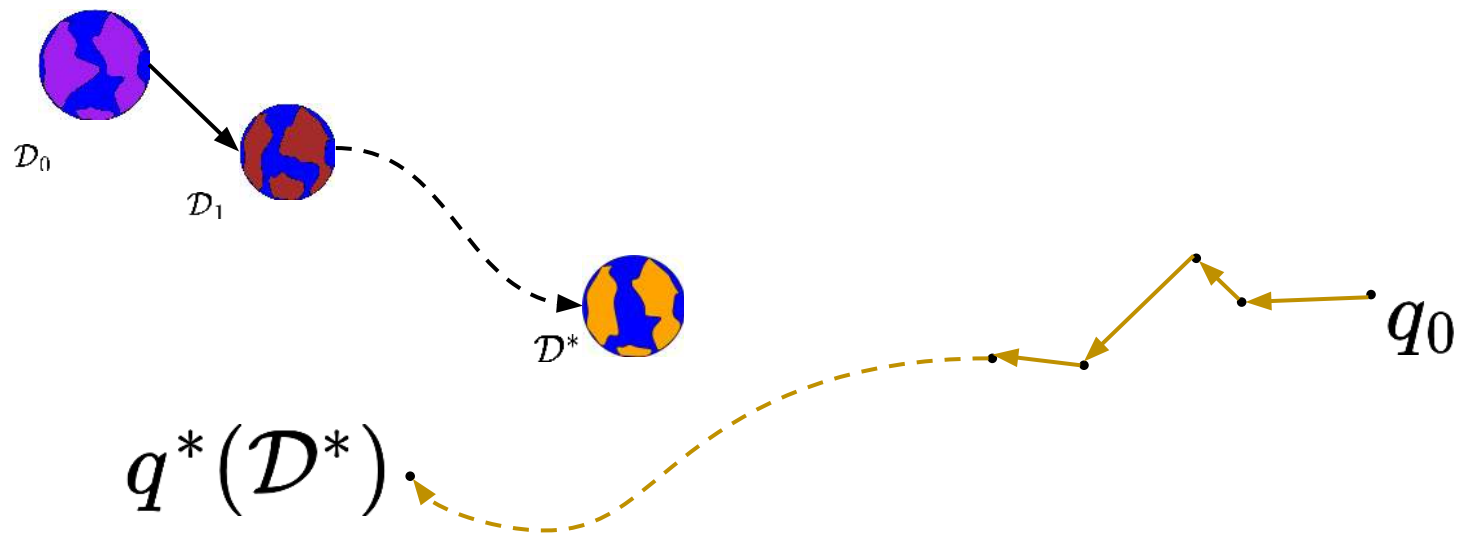




# Hierarchical reinforcement learning

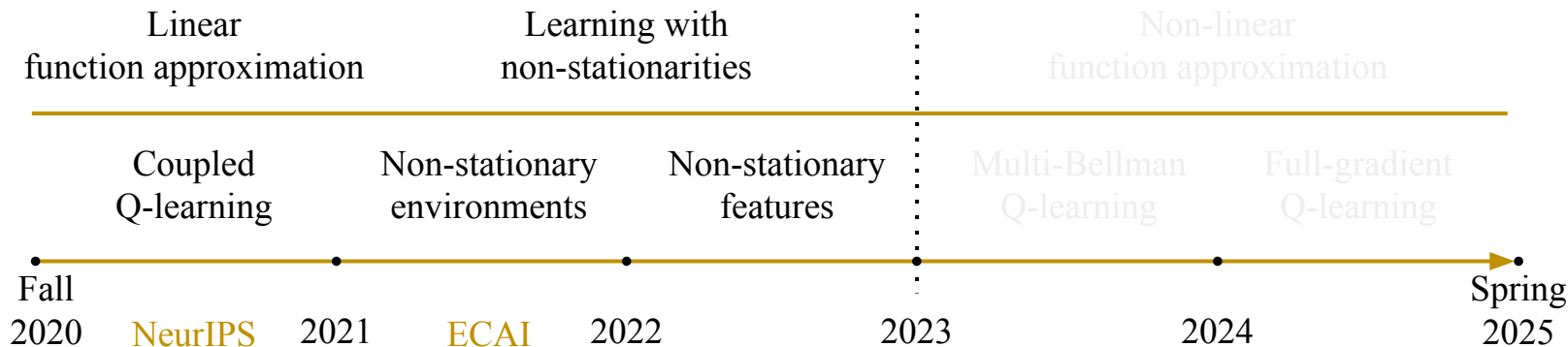


# Non-stationary environments



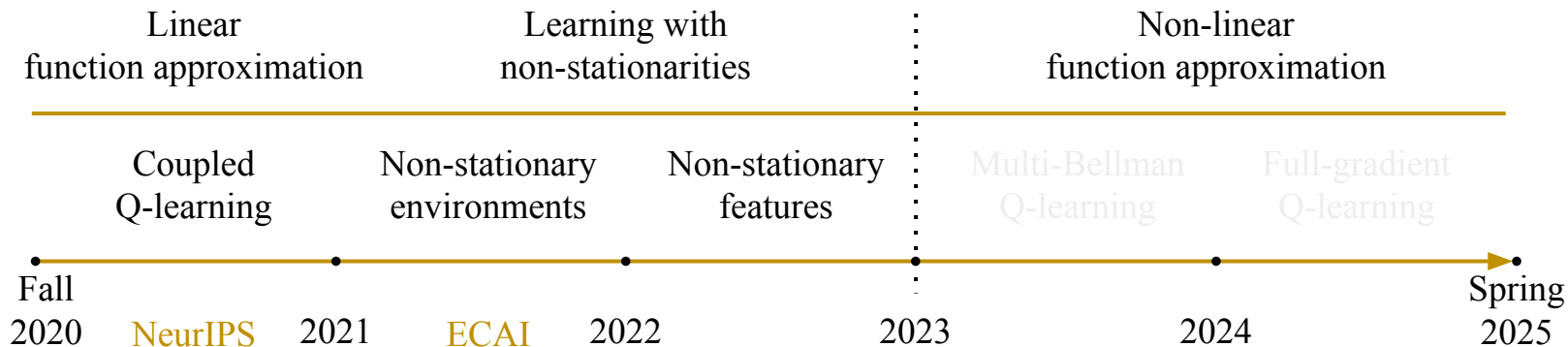
# Expected contributions

How can we approximate the optimal value function?



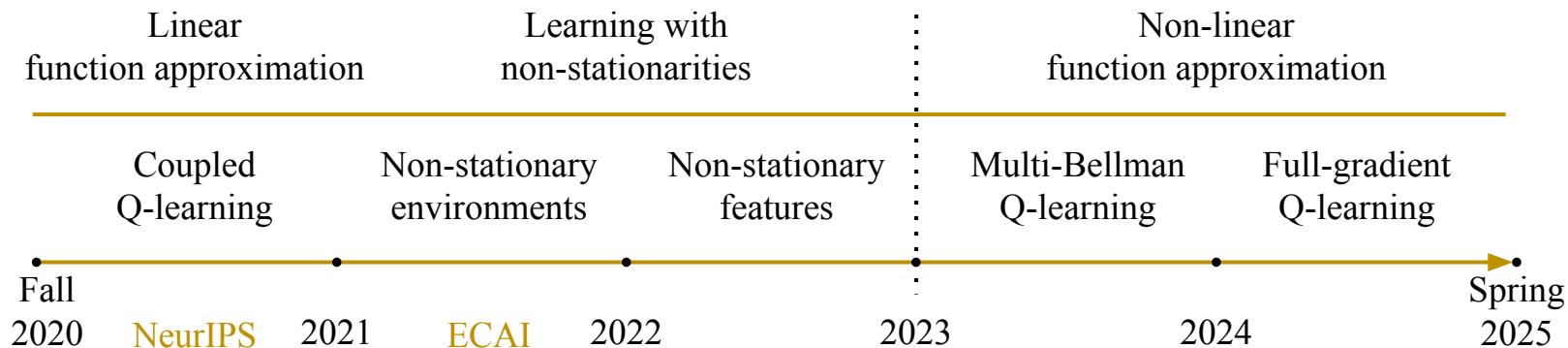
# Expected contributions

How can we approximate the optimal value function?

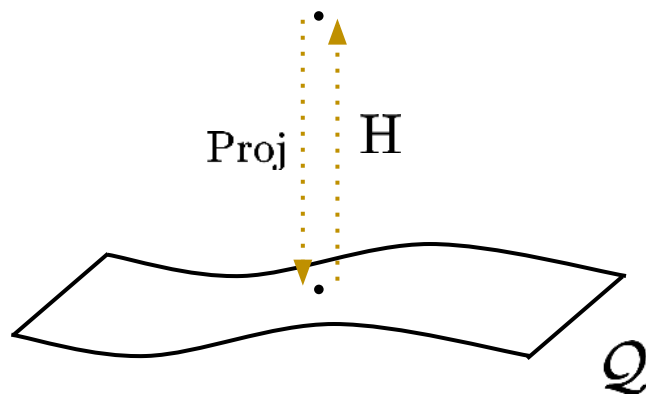


# Expected contributions

How can we approximate the optimal value function?

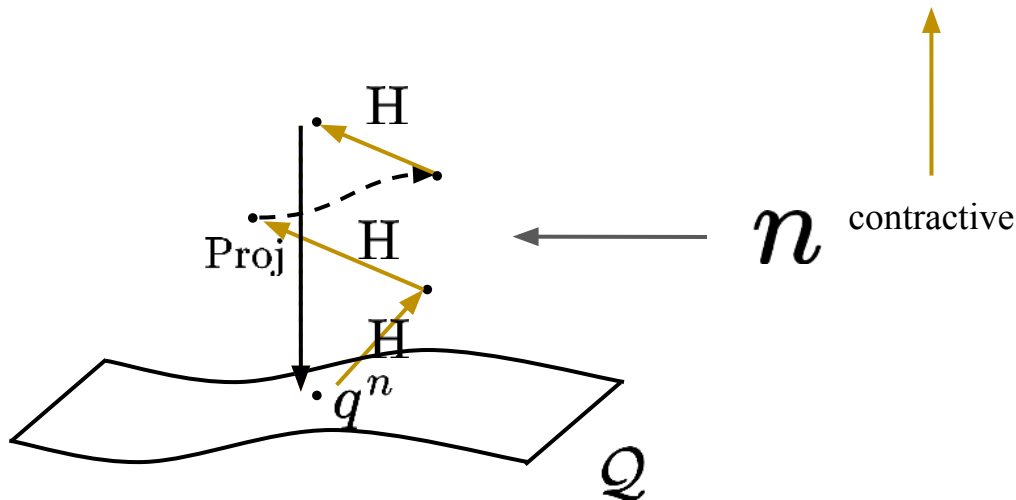


# Multi-Bellman Q-learning



# Multi-Bellman Q-learning

$$q^n = (\text{Proj } H^n) q^n$$



$$(H^{n+1}q)(x, a) = \mathbb{E} [r(x, a) + \gamma \cdot \max_{a'} H^n(x', a')]$$

## Multi-Bellman Q-learning

$$q^n = (\text{Proj } H^n) q^n$$

$$\max_i \mathbb{E} [Z_i] \neq \mathbb{E} [\max_i Z_i]$$

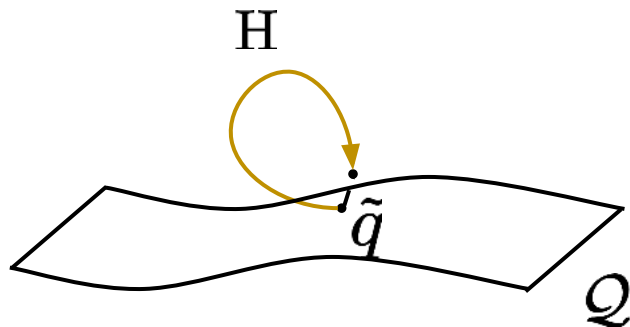
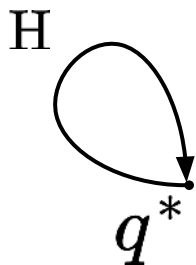
maximization of  
expected values

multi-sampling

$$(H^{n+1}q)(x, a) = \mathbb{E} [r(x, a) + \gamma \cdot \max_{a'} H^n(x', a')]$$

# Full-gradient Q-learning

$$\tilde{q} = \operatorname{argmin}_{q \in \mathcal{Q}} \|\mathbf{H}q - q\|$$

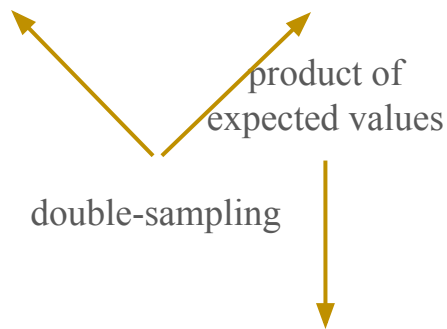


$$(\nabla l)(q) = \mathbb{E} [((\mathbf{H}q)(\mathbf{x}, \mathbf{a}) - q(\mathbf{x}, \mathbf{a})) \cdot ((\nabla (\mathbf{H}q))(\mathbf{x}, \mathbf{a}) - (\nabla q)(\mathbf{x}, \mathbf{a}))]$$

Full-gradient Q-learning

$$\tilde{q} = \operatorname{argmin}_{q \in \mathcal{Q}} \|\mathbf{H}q - q\|$$

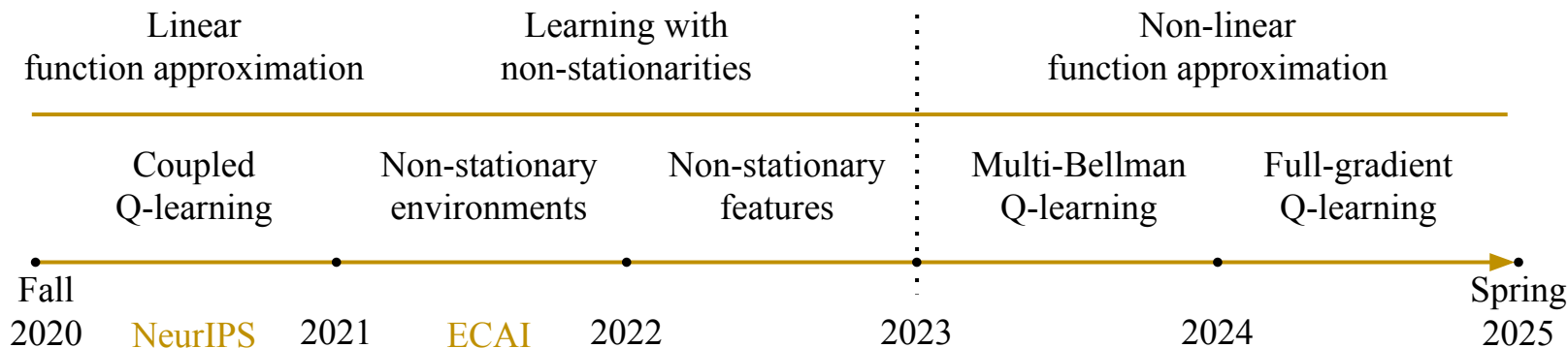
$$\mathbb{E}[f(\mathbf{Z})] \cdot \mathbb{E}[g(\mathbf{Z})] \neq \mathbb{E}[f(\mathbf{Z}) \cdot g(\mathbf{Z})]$$



$$(\nabla l)(q) = \mathbb{E} [((\mathbf{H}q)(\mathbf{x}, \mathbf{a}) - q(\mathbf{x}, \mathbf{a})) \cdot ((\nabla (\mathbf{H}q))(\mathbf{x}, \mathbf{a}) - (\nabla q)(\mathbf{x}, \mathbf{a}))]$$

# Expected contributions

How can we approximate the optimal value function?



Reinforcement learning



Function approximation



Research problem



Expected contributions



Discussion

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